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TAMIL NADU CENTRE OF EXCELLENCE FOR ADVANCED MANUFACTURING (TANCAM)

TNWISE2026

HACKATHON FOR WOMEN IN SCIENCE & ENGINEERING

THEME: Soil, Land & Crop Resource Engineering

TITLE: SoilSense Thozhan

COLLEGE: Loyola-ICAM College of Engineering and Technology

TEAM LEADER NAME: Mathalan Mary Delphin W

DASSAULT SYSTEMES

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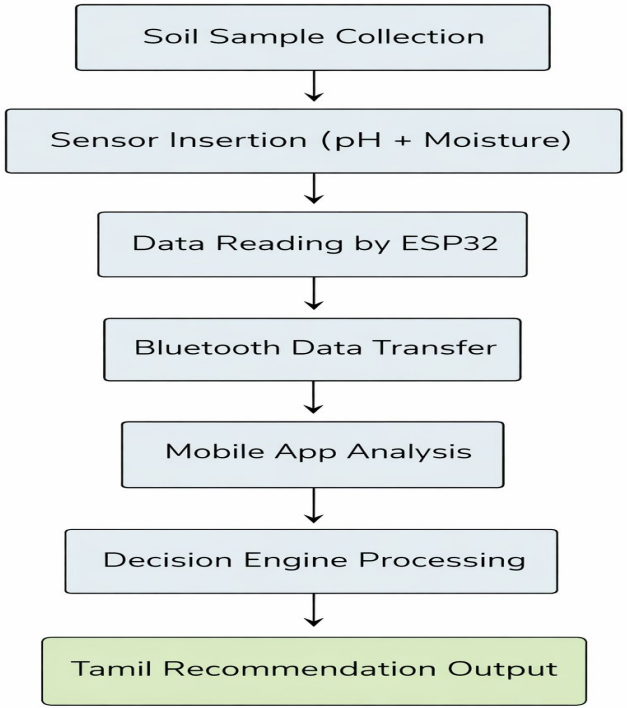
PROBLEM STATEMENT:

OBJECTIVES OF THE WORK:

- Small and marginal farmers lack access to affordable soil testing facilities.
- Laboratory soil testing is time-consuming, costly, and not easily accessible in rural areas.
- Even when soil reports are available, farmers struggle to interpret pH and nutrient values.
- Incorrect fertilizer usage due to lack of guidance leads to soil degradation and financial loss.
- There is a need for a low-cost, easy-to-use soil diagnostic device with localized decision support.



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PROJECT DESCRIPTION:

SoilSense Thozhan is a low-cost embedded soil sensing and decision support system designed for small and marginal farmers.

The system measures key soil parameters such as pH, moisture, and temperature using integrated sensors connected to an ESP32 module.

The collected data is transmitted to a mobile application via Bluetooth for real-time analysis.

A built-in decision engine converts soil data into simple, actionable fertilizer and crop recommendations in Tamil language.

CONCLUSION:

- The proposed system bridges the gap between soil testing and practical farming decisions.
- It reduces fertilizer misuse and helps farmers save input costs.
- The device is affordable, portable, and easy to use for rural and women farmers.
- SoilSense Thozhi has strong potential for village-level deployment through SHGs and agricultural extension programs.

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TEAM MEMBER DETAILS:

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TEAM MEMBER 3:

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TEAM MENTOR NAME:

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